INCH-POUND

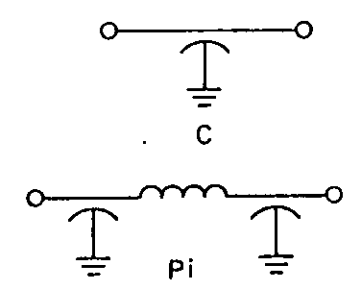
MIL-F-28861/10D
2 June 1989
SUPERSEDING
MIL-F-28861/10C
2 November 1988

MILITARY SPECIFICATION SHEET

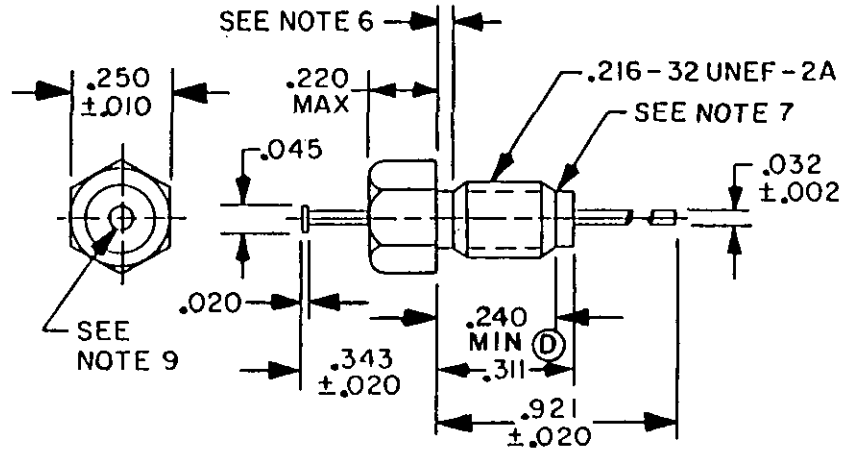
FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLE FS64

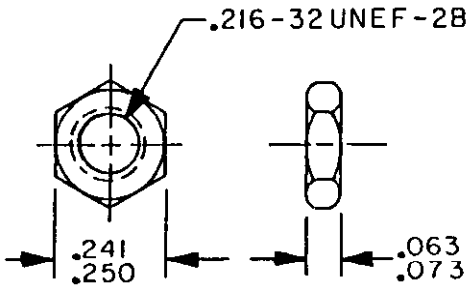
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.

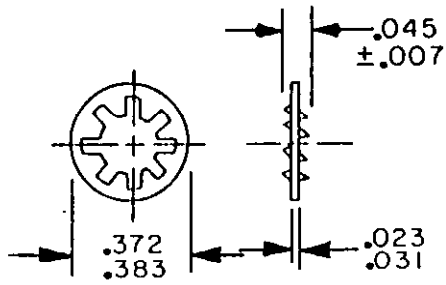


CIRCUIT DIAGRAMS





HEX NUT



LOCKWASHER

MOUNTING HARDWARE

FIGURE 1. Case and hardware dimensions and circuit diagrams.

Ⓓ denotes changes

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Inches	mm	Inches	mm
.002	0.05	.073	1.85
.007	0.18	.216	5.49
.010	0.25	.220	5.59
.020	0.38	.241	6.12
.023	0.58	.250	6.35
.031	0.79	.311	7.90
.032	0.81	.343	8.71
.045	1.14	.372	9.45
.063	1.60	.383	9.73
		.921	23.39

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Circuit diagrams are for information only.
5. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
6. Imperfect thread or undercut optional, .040 inch (1.02 mm) maximum.
7. One imperfect thread allowed .040 inch (1.02 mm) maximum.
8. Recommended mounting torque: 64 oz-in ±4 oz-in.
9. Turret head terminal optional.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 2.5 grams maximum.

Case finish:

Class B filters, S only (silver plated).
Classes A and S filters, S or G only (silver plated or gold plated). In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 10 amperes maximum.

Capacitance to ground: See table I for capacitance value.

Dissipation factor: 3.0 percent maximum.

Voltage and temperature limits of capacitance: +15 percent, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohm-microfarads or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarads or 10,000 megohms minimum, whichever is less.

Insertion loss:

At +25°C: In accordance with table I.

At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: 0.1 volt maximum.

DC resistance: 0.01 ohm maximum.

Seal: In accordance with MIL-F-28861. Leakage rate for classes A and S filters shall not exceed 1×10^{-7} atm cm^3/s.

Temperature rise: +25°C maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Quality assurance provisions:

Class B and class S filters: In accordance with MIL-F-28861.

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Class A filters:

Qualification inspection: Initial qualification may be granted based on qualification to class B. Baseline documentation in accordance with MIL-F-28861 is required and shall be submitted to the qualifying activity prior to completion of quality conformance inspection on the initial lot. Retention of qualification shall be in accordance with class S requirements.

In-process inspection: In accordance with class S requirements, except that the requirements of MIL-C-123 shall not apply.

Quality conformance inspection: The inspection lot definition and inspection of product for delivery shall be in accordance with the class S requirements.

Ⓓ TABLE I. Electrical characteristics.

Dash number	Circuit	Product assurance level			Rated voltage		Capacitance (µF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
		Class			DC volts	AC volts 3/		1 MHz	10 MHz	100 MHz	200 MHz	1 GHz	10 GHz
		A	B	S									
001	Pi	---	X	---	50	---	.12	20	38	70	70	70	70
002	C	---	X	---	50	---	.20	26	39	52	60	70	70
003	C	X	X	X	100	---	.05	15	35	42	50	70	70
004	C	X	X	X	100	---	.10	20	38	48	53	70	70
005	Pi	---	X	---	100	---	.075	18	38	70	70	70	70
006	C	X	X	X	200	115	1,000 pF	---	4	20	25	40	50
007	C	X	X	X	200	115	5,000 pF	---	15	34	41	50	55
008	C	X	X	X	200	115	.01	4	21	35	40	55	60
009	Pi	X	X	X	200	---	.012	---	12	60	70	70	70
010	C	X	X	X	200	115	.025	8	28	36	44	64	70

1/ For C circuits, insertion loss measurements shall be made under no load. For Pi circuits, insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz; above this frequency range, insertion loss measurements shall be made under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

3/ 0 to 400 Hz.

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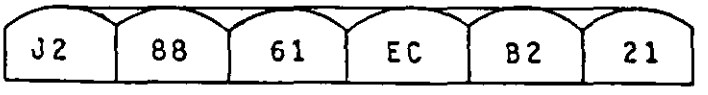
Product assurance level: See table I.

Part or Identifying Number (PIN): The Part or Identifying Number shall be as follows:

M28861/10 - XXX X X

- XXX: Dash no. per figure 1 and table I
- X: Case finish
- X: Product assurance level per MIL-F-28861 and table I

Marking: Filters shall be marked, as a minimum, with an abbreviated Part or Identifying Number as shown on figure 2. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.



J - Abbreviation for JAN
28861 - Abbreviation for M28861
E - Indicates specification sheet 10
C - Code indicating dash no. as follows:

Code	Dash no.	Code	Dash no.
A	001	F	006
B	002	G	007
C	003	H	008
D	004	J	009
E	005	K	010

B - Code indicating product assurance level (for example, B indicates class B).
221 - Date code. First digit indicates the year (for example, 2 indicates 1982). The last two digits indicate the week.

FIGURE 2. Example of marking for the abbreviated Part or Identifying Number on the hex flats - expanded view.

Cataloging information: Dash numbers 002 through 004, 006 through 008, and 010 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 001, 005, and 009 shall be cataloged under FSC 5915 as radio frequency interference filters.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0081-05)

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